

Program : Diploma in Wood and Paper Technology	
Course Code : 3127	Course Title: Pulp Technology Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To familiarize the students with Pulp testing
- To familiarize students with recovery of chemicals used in pulping processes with due techno-economic and environmental considerations

Course Prerequisites:

Topic	Course code	Course name	Semester
Pulping		Pulp technology	3
Chemistry of Wood		Wood chemistry	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Conduct the analysis of white liquor	12	Applying
CO2	Perform the analysis of bleach liquor	12	Applying
CO3	Analyse the Black liquor	6	Applying
CO4	Demonstrate the analysis of pulp	12	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Conduct the analysis of white liquor		
M1.01	Demonstrate the Preparation of Pulping liquor	3	Applying
M1.02	Determine the percentage of NaOH, Na ₂ S, Na ₂ CO ₃ , Na ₂ SO ₄	3	Applying
M1.03	Determine the percentage of TTA, TAA, sulphidity& causticizing efficiency	6	Applying
CO2	Perform the analysis of bleach liquor		
M2.01	Demonstrate the preparation of bleach liquor	3	Applying
M2.02	Determine pH, total available chlorine, strength of caustic soda	3	Applying
M2.03	Determine percentage of sodium hydroxide, sodium silicate magnesium sulphate	6	Applying
	Lab Exam – I	1.5	
CO3	Analyse the black liquor		
M3.01	Determine the PH, acidity, alkalinity of black liquor	3	Applying
M3.02	Determine total solids and total alkali	3	Applying
CO4	Demonstrate the analysis of pulp		
M4.01	Prepare pulp from waste paper and determine the consistency of pulp	3	Applying
M4.02	Determine kappa number and permanganate number	3	Applying

M4.03	Determine the ash content	3	Applying
M4.04	Demonstrate the identification and microscopic study of various types of pulp fibres'- reed, bamboo, eucali hybrid	3	Applying
	Lab Exam – II	1.5	

Text / Reference:

T/R	Book Title/Author
T ₁	K.W.Britt, Hand book of Pulp and paper technology, C.B.S.Publishers, New Delhi
R ₁	Tappi Test Methods for pulp testing, T 1-200 series
R ₂	Tappi test Methods for paper Testing, T 400-500